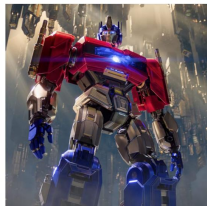
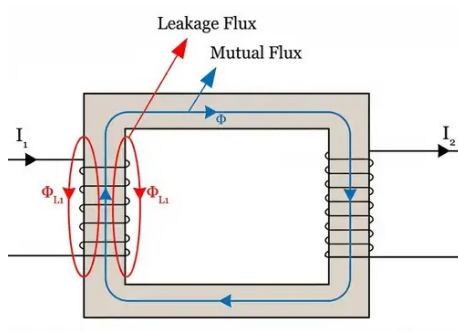


- Mutual inductance of coupled coils, Dot convention
- Ideal Transformer
- Equivalent circuit and non-ideal transformer
- OC-SC Tests
- Voltage Regulation, Efficiency
- Skin Effect

Transformers



Mutual and Leakage Flux



Quick recap!

- If a current I_1 flows in a coil, it induces a magnetic flux ϕ_{11} through itself.
- The amount of flux produced is proportional to the amount of current flowing through the coil.
- We define the **self-inductance** of the coil as this constant of proportionality -

$$L_1 = \frac{N_1 \phi_{11}}{I_1}$$

Mutual Inductance

Consider two coils wound on the same core. Suppose the coil 2 carries current I_2 . Current I_2 induces a magnetic flux ϕ_{22} through the coil 2. Some part of the flux ϕ_{22} also passes through the coil 1. Let us call it ϕ_{12} (i.e, the flux in coil 1 due to current in coil 2).

Similarly, suppose coil 1 carries current I_1 . This current I_1 induces a magnetic flux ϕ_{11} through the coil 1. Some part of the flux ϕ_{11} also passes through the coil 2. Let us call it ϕ_{21} (i.e, the flux in coil 2 due to current in coil 1).

We define the **mutual-inductance** of the coils 1 and 2 as -

$$M = \frac{N_1 \phi_{12}}{I_2} = \frac{N_2 \phi_{21}}{I_1}$$

Coupling Coefficient

Assume that a fraction K of the flux ϕ_{22} (total flux due to current I_2 in coil 2) passes through coil 1, and the rest $(1 - K)$ leaks out.

$$\phi_{12} = K\phi_{22}$$

$$M_{12} = \frac{N_1\phi_{12}}{I_2} = \frac{N_1K\phi_{22}}{I_2} = K \frac{N_1}{N_2} \cdot \underbrace{\frac{N_2\phi_{22}}{I_2}}_{L_2} = K \frac{N_1}{N_2} L_2$$

Assuming uniform cross-section, we have $\frac{N_1}{N_2} = \sqrt{\frac{L_1}{L_2}}$

$$\therefore \boxed{M = K\sqrt{L_1L_2}}$$

We define K as the **coefficient of coupling** ($0 \leq K \leq 1$)

Magnetic energy stored in coupled coils

Magnetic energy stored in coupled coils is given by -

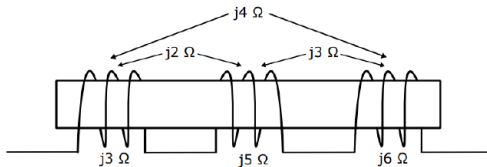
$$E = \frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2 \pm MI_1I_2$$

Take +ve sign if both currents enter dotted terminal or both currents enter undotted terminal

Tutorial Problem

Obtain the dotted equivalent for the circuit shown below and use the equivalent to find the equivalent inductive reactance.

Answer: An inductive reactance of $12\ \Omega$.



Dot Convention

Rule for **self-induced emf**

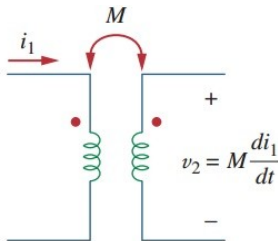
Voltage **drops** by $L_2 \frac{di_2}{dt}$ along chosen direction of current i_2
(Just like a good ol' inductor)

Dot Convention

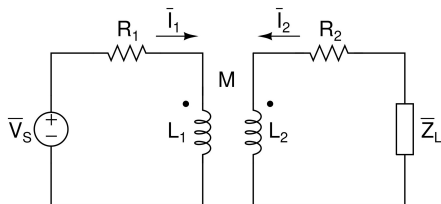
Rule for **mutual-induced emf**

In order to find mutually-induced emf in coil 2 due to current in coil 1 -

- If the chosen direction of I_1 is **entering** the dot in coil 1 $\implies$
Mutual-induced emf in coil 2 is $M \frac{dI_1}{dt}$ with *positive* polarity at the dot on coil 2
- If the chosen direction of I_1 is **leaving** the dot in coil 1 $\implies$
Mutual-induced emf in coil 2 is $M \frac{dI_1}{dt}$ with *negative* polarity at the dot on coil 2



Analysis of coupled coils



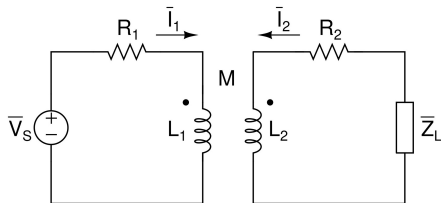
Time-domain equations -

$$\text{Voltage across coil 1} = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$\text{Voltage across coil 2} = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

Check that all sign conventions have been followed correctly!

Phasor analysis of coupled coils



Phasor form -

$$\bar{E}_1 = j\omega L_1 \bar{I}_1 + j\omega M \bar{I}_2$$

$$\bar{E}_2 = j\omega L_2 \bar{I}_2 + j\omega M \bar{I}_1$$

Phasor analysis of coupled coils

Applying KVL on secondary side,

$$(j\omega L_2 + R_2 + \overline{Z_L})\overline{I_2} + j\omega M\overline{I_1} = 0$$

$$\therefore \overline{I_2} = \frac{-j\omega M\overline{I_1}}{j\omega L_2 + R_2 + \overline{Z_L}}$$

Applying KVL on primary side,

$$\overline{V_s} = (R_1 + j\omega L_1)\overline{I_1} + j\omega M\overline{I_2}$$

Substitute $\overline{I_2}$ from above expression and simplify to get...

Reflected Impedance

$$\frac{\overline{V_s}}{\overline{I_1}} = \overline{Z_{in}} = \overline{Z_{11}} + \frac{\omega^2 M^2}{\overline{Z_{22}}}$$

Where:

$\overline{Z_{11}} = R_1 + j\omega L_1$ (impedance on primary side in isolation)

$\overline{Z_{22}} = R_2 + j\omega L_2 + \overline{Z_L}$ (impedance on secondary side in isolation)

$\frac{\omega^2 M^2}{\overline{Z_{22}}}$ is called **reflected impedance**

Ideal Transformer

- No winding losses $\implies R_1 = R_2 = 0$
- No leakage flux $\implies M = \sqrt{L_1 L_2}$
- Core made of perfect magnetic material $\implies$ No core losses and $\mathcal{R} = 0$ which implies $L_1 \rightarrow \infty$ and $L_2 \rightarrow \infty$

Input impedance of an Ideal Transformer

Define the turns ratio,

$$a = \frac{N_2}{N_1} = \frac{L_2^2}{L_1^2}$$

$$M^2 = L_1 L_2 = a^2 L_1^2$$

$$\text{Input impedance, } \overline{Z}_{in} = \overline{Z}_{11} + \frac{\omega^2 M^2}{\overline{Z}_{22}} = j\omega L_1 + \frac{\omega^2 a^2 L_1^2}{j\omega a^2 L_1 + \overline{Z}_L}$$

Simplify to get,

$$\overline{Z}_{in} = \frac{\overline{Z}_L}{\frac{\overline{Z}_L}{j\omega L_1} + a^2}$$

Take limit $L_1 \rightarrow \infty$ to get...

Ideal Transformer

Input impedance of an ideal transformer is given by -

$$\overline{Z}_{in} = \frac{\overline{Z}_L}{a^2}$$

This means that given any circuit containing an ideal transformer, we can reflect impedances from one side to other using this formula, remove the transformer and solve it like a simple circuit!

Also, since there is no leakage for an ideal transformer, the same flux ϕ links both the coils.

$$V_1 = N_1 \frac{d\phi}{dt} \quad \text{and} \quad V_2 = N_2 \frac{d\phi}{dt}$$

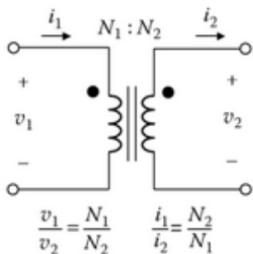
$$V_1 = \frac{V_2}{a}$$

Dot Convention for Ideal Transformer

Note that we can't write a KVL equation for an ideal transformer in terms of self and mutual inductances, like we did earlier for general coupled coils (because L and M are infinite for ideal transformer)

Dot Convention for Ideal Transformer

Therefore, for an ideal transformer, the dot convention is as follows -



Choose the dotted terminal as positive voltage on both sides.

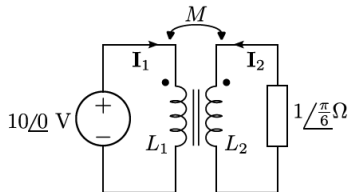
Choose current entering the dot on one side, and exiting the dot on the other side.

With these polarities chosen, $V_2/V_1 = N_2/N_1$ and $I_2/I_1 = N_1/N_2$ are satisfied.

If you make any of these chosen polarities opposite in the circuit, you need to put a minus sign in front of that quantity in the equation.

Tutorial Problem

Question 4:



This coupled magnetic circuit operates in the sinusoidal steady state at a frequency $\omega = 314.15 \text{ } rs^{-1}$. The inductances $L_1 = 100 \text{ mH}$, $L_2 = 50 \text{ mH}$ and the mutual inductance $M = \sqrt{L_1 L_2}$. The complex load impedance is $Z = 1/\angle \pi/6 \Omega$. Calculate the currents I_1 and I_2 . Work out the complex power balance of your solution.

Solution

Writing KVL for both loops,

$$10\angle 0^\circ - j\omega L_1 \bar{I}_1 - j\omega M \bar{I}_2 = 0 \quad \text{for primary side}$$

$$j\omega L_2 \bar{I}_2 + j\omega M \bar{I}_1 + \bar{I}_2 \times 1\angle 30^\circ = 0 \quad \text{for secondary side}$$

From the second equation, we get

$$\bar{I}_2 = \left(\frac{-j\omega M}{j\omega L_2 + 1\angle 30^\circ} \right) \bar{I}_1$$

$$\bar{I}_2 = (1.368\angle -176.9^\circ) \bar{I}_1$$

Substituting in first equation, we get

$$\bar{I}_1 = 5.166\angle -33^\circ \text{ A}$$

$$\bar{I}_2 = 7.067\angle 150^\circ \text{ A}$$

To check power balance,

$$\begin{aligned} S_{\text{source}} &= \bar{V}_1 \bar{I}_1^* = (10 \angle 0^\circ) (5.166 \angle +33^\circ) \\ &= 51.66 \angle +33^\circ \text{ VA} = (43.3 + 28.13j) \text{ VA} \end{aligned}$$

$$\begin{aligned} S_{\text{load}} &= \bar{V}_2 \bar{I}_2^* = (\bar{I}_2 \cdot Z) \bar{I}_2^* = |\bar{I}_2|^2 Z \\ &= (7.067)^2 (1 \angle 30^\circ) \\ &= 49.94 \angle 30^\circ \text{ VA} = (43.3 + 24.97j) \text{ VA} \end{aligned}$$

Why is there a mismatch!? Where else is power being dissipated?

Solution

The coupled coils dissipate reactive power!

The voltage across primary coil, and power dissipated in it will be

$$\bar{E}_1 = j\omega L_1 \bar{I}_1 + j\omega M \bar{I}_2, \quad \bar{S}_1 = \bar{E}_1 \bar{I}_1^*$$

Similarly voltage across secondary coil, and power in it will be

$$\bar{E}_2 = j\omega L_2 \bar{I}_2 + j\omega M \bar{I}_1, \quad \bar{S}_2 = \bar{E}_2 \bar{I}_2^*$$

Total complex power dissipated in coils is

$$\begin{aligned} \bar{S}_{\text{coil}} = \bar{S}_1 + \bar{S}_2 &= j\omega L_1 |\bar{I}_1|^2 + j\omega M \bar{I}_2 \bar{I}_1^* + j\omega L_2 |\bar{I}_2|^2 + j\omega M \bar{I}_1 \bar{I}_2^* \\ &= j3.118 \text{ VA} \end{aligned}$$

We note that

$$\bar{S}_{\text{source}} = \bar{S}_{\text{coil}} + \bar{S}_{\text{load}}$$

Power is balanced :)

Tutorial Problem

Question 5:

Given $L_1 = 0.2$ H, $L_2 = 0.1$ H, $M = 0.1$ H, and $R = 10\ \Omega$ in the circuit below, find i_1 for $v_1 = 142.3 \sin 100t$.

Answer: $i_1 = 9 \sin (100t - 71.6^\circ)$ A.

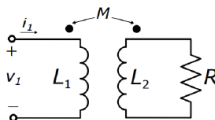


Fig. 1. Diagram for Question 1

Solution

Write KVL, (Take $\bar{I}_2$ to be entering the dot)

$$\bar{V}_1 - j\omega L_1 \bar{I}_1 - j\omega M \bar{I}_2 = 0 \quad (1)$$

$$j\omega L_2 \bar{I}_2 + j\omega M \bar{I}_1 + \bar{I}_2 R = 0 \quad (2)$$

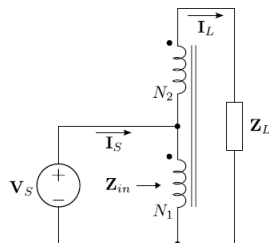
From (2), you get

$$\bar{I}_2 = \left(\frac{-j\omega M}{j\omega L_2 + R} \right) \bar{I}_1 = (0.707 \angle -135^\circ) \bar{I}_1$$

$$\begin{aligned} \therefore \bar{I}_1 &= \frac{\bar{V}_1}{j\omega L_1 + j\omega M(0.707 \angle -135^\circ)} \\ &= 9 \angle -71.6^\circ \text{ A} \end{aligned}$$

$$\therefore i_1(t) = 9 \sin(100t - 71.6^\circ) \text{ A}$$

Tutorial Problem



This question is on an *autotransformer* connection. The two windings of an ideal transformer are electrically interconnected as shown. (An ideal transformer has no leakage magnetic flux, no internal power loss, and self-inductances $(L_1, L_2) \rightarrow \infty$.) Note the winding polarities. $N_1 = 1000$ and $N_2 = 100$ turns. $V_S = 100\angle 0^\circ \text{ V}$. The load impedance $Z_L = 1/\pi/6 \, \Omega$. What is the input impedance Z_{in} seen by the voltage source? What is the complex power S_L taken by the load? What is the complex power S_S given out by the voltage source V_S ? Repeat the question with the polarity of Winding 2 reversed.

Answer: $Z_{in} = V_S/I_S = 0.826/\pi/6 \, \Omega$.

Solution

Following the dot convention of an ideal transformer,

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{1}{10}$$

$$\therefore V_2 = \frac{V_1}{10} = \frac{100\angle 0^\circ}{10} = 10\angle 0^\circ \text{ V}$$

$\therefore$ Voltage across $\bar{Z}_L$ is $V_1 + V_2 = 110\angle 0^\circ \text{ V}$

$$\therefore \bar{I}_L = \frac{110\angle 0^\circ}{\bar{Z}_L} = \frac{110\angle 0^\circ}{1\angle 30^\circ} = 110\angle -30^\circ \text{ A}$$

Solution

$$\frac{I_2}{I_1} = \frac{N_1}{N_2} = 10 \Rightarrow \frac{\bar{I}_L}{\bar{I}_S - \bar{I}_L} = 10 \quad \text{which gives} \quad \bar{I}_S = \frac{11}{10} \bar{I}_L$$

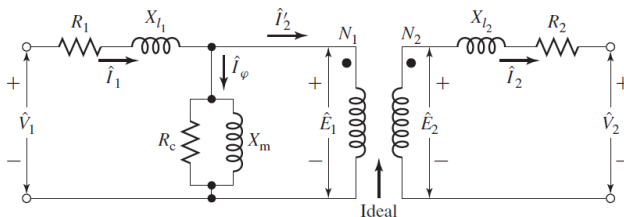
(Note that we chose $\bar{I}_2$ exiting the dot and $\bar{I}_1$ entering the dot in accordance with the dot convention)

$$\therefore \bar{I}_S = 121 \angle -30^\circ \text{ A}$$

$$\therefore \bar{Z}_{in} = \frac{\bar{V}_S}{\bar{I}_S} = \frac{100 \angle 0^\circ}{121 \angle -30^\circ} = 0.826 \angle +30^\circ \Omega$$

Equivalent circuit of Non-Ideal Transformer

(Check out the derivation of the circuit from the lecture slides - Important!)



In this equivalent circuit, since there is an ideal transformer in the middle, we can reflect the impedances from primary to secondary or vice versa.

Equivalent Circuit Parameters

- R_1 and R_2 are the **winding resistances**
- X_{l1} and X_{l2} are the **leakage reactances**
These represent the inductance due to the leakage flux
- X_m is the **magnetizing reactance**
It represents the inductance due the mutual flux in the magnetic core
- R_c is the **core loss resistance**
It represents energy losses in the core (hysteresis and eddy-current losses)

Note that for an **ideal transformer**, $R_1 = R_2 = 0$, there is no leakage so $X_{l1} = X_{l2} = 0$, core is a perfect magnetic material so its inductance is $X_m \rightarrow \infty$, and there are no core losses so $R_c \rightarrow \infty$. Putting these values in the equivalent circuit, we are left with nothing but the ideal transformer as expected!

Short Circuit Test

The **short circuit test** is performed to find out the **equivalent series impedance** $R_{eq} + jX_{eq}$

where $R_{eq} = R_1 + R'_2$ and $X_{eq} = X_{l1} + X'_{l2}$ (shunt branch is ignored).

Short the output side of the transformer ($\overline{Z_L} = 0$)

Reflect R_2 and X_{l2} to the primary side (R'_2 and X'_{l2})

Apply $\overline{V_{SC}}$ at input of the transformer, and measure the primary current drawn $\overline{I_{SC}}$ and power consumed P

$$R_{eq} = \frac{P}{I_{SC}^2}$$

$$X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

$$\text{where } Z_{eq} = \frac{V_{SC}}{I_{SC}} = \left| \frac{\overline{V_{SC}}}{\overline{I_{SC}}} \right|$$

Open Circuit Test

The **open circuit test** is performed to find out the **core parameters** X_m and R_c

Open the output side of the transformer ($\overline{Z_L} = \infty$)

Apply $\overline{V_{OC}}$ at input of the transformer, and measure the primary current drawn $\overline{I_{OC}}$ and power consumed P

$$\therefore I_2 = I'_2 = 0$$

Hence, current drawn from source (I_{OC}) = current going into shunt branch

$$R_c = \frac{V_{OC}^2}{P}$$

$$X_m = \frac{1}{\sqrt{(1/Z_{OC})^2 - (1/R_c)^2}}$$

$$\text{where } Z_{OC} = \frac{V_{OC}}{I_{OC}} = \left| \frac{\overline{V_{OC}}}{\overline{I_{OC}}} \right|$$

Tutorial Problem

A 20 kVA, 200/1000 V, single-phase transformer gave the following test results:

- O.C. Test (LV side): 200 V, 1.2 A, 90 W
- S.C. Test (HV side): 50 V, 5 A, 110 W

Calculate the parameters of the equivalent circuit referred to the LV side.

Answer: Referred to LV side: $(R'_1 + R_2) = 0.176 \Omega$, $(X'_1 + X_2) = 0.3592 \Omega$

Solution

From SC Test,

$$(R_{eq})_{HV} = \frac{P_{sc}}{I_{sc}^2} = \frac{110}{5^2} = 4.4 \Omega$$

$$(Z_{eq})_{HV} = \frac{V_{sc}}{I_{sc}} = \frac{50}{5} = 10 \Omega$$

$$(X_{eq})_{HV} = \sqrt{Z_{eqHV}^2 - R_{eqHV}^2} = 8.98 \Omega$$

Referred to LV side, we get

$$(R_{eq})_{LV} = \frac{(R_{eq})_{HV}}{a^2} = 0.176 \Omega$$

$$(X_{eq})_{LV} = \frac{(X_{eq})_{HV}}{a^2} = 0.3592 \Omega$$

Solution

From OC Test, (on LV side)

$$R_c = \frac{V_{oc}^2}{P_{oc}} = \frac{200^2}{90} = 444.44 \, \Omega$$

$$Z_{oc} = \frac{V_{oc}}{I_{oc}} = \frac{200}{1.2} = 166.67 \, \Omega$$

$$X_m = \frac{1}{\sqrt{\left(\frac{1}{Z_{oc}}\right)^2 - \left(\frac{1}{R_c}\right)^2}} = 179.79 \, \Omega$$

We get

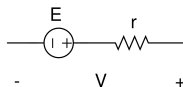
$$(R_c)_{LV} = 444.44 \, \Omega \quad (X_m)_{LV} = 179.79 \, \Omega$$

Voltage Regulation

Before looking at voltage regulation of transformers, let us try to understand what is 'voltage regulation' in general.

Recall that we model a real battery (or a voltage source) as an ideal battery E along with a series internal resistance r . This means that for a real battery, the **terminal voltage** V decreases as the current drawn from the battery increases.

$$V = E - Ir$$



We want the battery to be able to supply large current without significant attenuation in terminal voltage V .

Voltage regulation is a measure of how much the terminal voltage drops with increase in current drawn.

Voltage Regulation

Voltage regulation is defined as the drop in terminal voltage upon increasing load from no-load to full(rated)-load, divided by no-load terminal voltage. It is expressed as a percentage.

$$\text{Voltage regulation} = \frac{V_{(OC)} - V_{(\text{full-load})}}{V_{(OC)}} \times 100\%$$

where $V_{(OC)}$ is the terminal voltage at no-load (open-circuit) conditions, $V_{(\text{full-load})}$ is the terminal voltage at full(rated)-load conditions

Remember - In power engineering, '**load**' refers to **load current**

Voltage regulation of Transformer

Considering the secondary coil output of a transformer as a voltage source, we can find the voltage regulation of the transformer.

$$\text{Voltage Regulation} \approx \left(\frac{I_2 R_{eq2} \cos \theta_2 + I_2 X_{eq2} \sin \theta_2}{V_{2(OC)}} \right) \times 100\%$$

where -

I_2 is the rated load current,

$R_{eq2} = R'_1 + R_2$ and $X_{eq2} = X'_1 + X_2$,

θ_2 is the power factor angle of the load impedance,

$V_{2(OC)} = aV_1$ is the no-load terminal voltage

(Note that at no-load condition, since current in secondary is zero, the terminal voltage is simply the input voltage V_1 stepped up/down by a - Verify yourself from the equivalent circuit diagram!)

Voltage Regulation of Transformer

$$\text{Voltage Regulation} \approx \left(\frac{I_2 R_{eq2} \cos \theta_2 + I_2 X_{eq2} \sin \theta_2}{V_{2(OC)}} \right) \times 100\%$$

Important -

Take θ_2 positive for **inductive load**

Take θ_2 negative for **capacitive load**

Losses in a Transformer

There are two types of **losses** in a transformer:

- **Coil loss (Copper loss)** — due to resistance of the windings

$$\text{Copper loss, } P_{\text{copper}} = I_2^2 R_{eq2}$$

- **Core loss (Iron loss)** — due to hysteresis and eddy currents in the magnetic core

$$\text{Iron loss, } P_{\text{iron}} = \frac{V_1^2}{R_c}$$

Important -

Iron loss is the power consumed in **open-circuit test**!

$$P_{\text{iron}} = P_{OC}$$

Copper loss is the power consumed in **short-circuit test**!

$$P_{\text{copper}} = P_{SC}$$

Efficiency of Transformer

$$\text{Efficiency, } \eta = \frac{V_2 I_2 \cos \theta_2}{V_2 I_2 \cos \theta_2 + P_{\text{iron}} + P_{\text{copper}}}$$

Condition for maximum efficiency -

Set $\frac{d\eta}{dI_2} = 0$ and solve to get -

$$P_{\text{iron}} = I_2^2 R_{eq2}$$

How to tackle questions on transformers?

Questions on transformers typically mention the ratings of the transformer.
Let us unpack the meaning of each rating -

For example, consider a -

100kVA, 440/220 V, 1-phase, 50Hz transformer ...

- 100kVA is the **rated output apparent power**, $V_2 I_2 = 100\text{kVA}$
- 440/220 V indicates that the **turns ratio** of the transformer is $a = N_2/N_1 = V_2/V_1 = 220/440$, hence $a = 0.5$
- 440/220 V also indicates that at rated load condition, the output voltage V_2 is 220V, therefore **rated load current** will be $I_2 = 100\text{kVA}/220\text{V} = 454.5\text{A}$
- 50Hz is the **frequency** (obviously)

Tip - When the question says that the transformer is running at rated load (or full load), that means that $V_2 I_2 =$ the given apparent power rating!

Tutorial Problem

A 100 kVA, 440/220 V, 1-phase, 50 Hz, core type transformer has an efficiency of 98.5% when supplying a full (rated) load at 0.8 power factor lag and an efficiency of 99%, when supplying half the full load at unity power factor. Find the iron (no-load) losses and copper (winding or load) losses under rated load conditions.

Answer: $w_i = 267.3 \text{ W}$; $w_c = 950.9 \text{ W}$

Solution

Since it isn't mentioned explicitly in the question, we assume $V_1 = 440 \text{ V}$ and $V_2 = 220 \text{ V}$ are not changed.

'Full load' and 'Half load' means that we are only changing output current by varying load impedance.

Therefore, since V_1 does not change, we assume that iron loss is constant.

Let $x = \text{iron-loss}$

$y = \text{copper-loss at rated (full) load}$

Note that copper loss $\propto I_2^2$

$$\therefore \text{Copper loss at half load} = \frac{y}{4}$$

Solution

Given, $V_2(I_2)_{\text{rated}} = 100 \text{ kVA}$

$$\therefore \eta_{\text{full}} = \frac{V_2(I_2)_{\text{rated}} \cos \theta_2}{V_2(I_2)_{\text{rated}} \cos \theta_2 + x + y}$$

$$\Rightarrow \frac{98.5}{100} = \frac{100 \times 10^3 \times 0.8}{100 \times 10^3 \times 0.8 + x + y} \Rightarrow x + y = 1218.27 \text{ W} \quad (1)$$

$$\eta_{\text{half}} = \frac{\frac{V_2(I_2)_{\text{rated}}}{2} \cos \theta_2}{\frac{V_2(I_2)_{\text{rated}}}{2} \cos \theta_2 + x + \frac{y}{4}}$$

$$\Rightarrow \frac{99}{100} = \frac{\frac{100 \times 10^3}{2} \times 1}{\frac{100 \times 10^3}{2} \times 1 + x + \frac{y}{4}} \Rightarrow x + \frac{y}{4} = 505.05 \text{ W} \quad (2)$$

Solving (1) and (2), we get $x = 267.3 \text{ W}$, $y = 950.9 \text{ W}$

Tutorial Problem

A 100 kVA, 1000/10000 V, 50 Hz transformer has an iron loss of 1200 W. The copper loss with 6 A in the high voltage winding is 500 W. If the output terminal voltage is maintained at 10000 V, find the load for maximum efficiency. Also, find the maximum efficiency at:

(a) unity power factor

(b) 0.8 power factor (lagging)

Answer: Load as fraction of the rated load at which maximum efficiency occurs: 0.9295

$$\eta = 97.48\% \text{ (upf)}$$

$$\eta = 96.87\% \text{ (0.8 pf)}$$

Solution

V_2 maintained at 10000 V $\Rightarrow V_1$ maintained at 1000 V

$\Rightarrow$ Iron loss, P_{Fe} remains 1200 W constant

Also, copper loss, $P_{Cu} \propto I_2^2$. For maximum efficiency,

$$P_{Cu} = P_{Fe} \Rightarrow \left(\frac{I_2^2}{6^2} \right) 500 = 1200 \Rightarrow I_2 = 9.29 \text{ A}$$

(a)

$$\eta = \frac{V_2 I_2 \cos \theta_2}{V_2 I_2 \cos \theta_2 + P_{cu} + P_{Fe}} = \frac{10000 \times 9.29 \times 1}{10000 \times 9.29 \times 1 + 1200 + 1200} = 97.48\%$$

(b)

$$\eta = \frac{V_2 I_2 \cos \theta_2}{V_2 I_2 \cos \theta_2 + P_{cu} + P_{Fe}} = \frac{10000 \times 9.29 \times 0.8}{10000 \times 9.29 \times 0.8 + 1200 + 1200} = 96.87\%$$

Last year's endsem question

A 10 kVA, 2500/250 V single-phase transformer has the following test results:

Open-circuit test: 250 V, 0.8 A, 50 W

Short-circuit test: 60 V, 3 A, 45 W

- (a) Calculate the efficiency η at 25% of full load at a power factor of 0.8.

$$\text{Short-circuit test} \Rightarrow 45 = 3^2 \times (R_{eq})_{HV} \Rightarrow (R_{eq})_{HV} = 5 \Omega$$

(done from HV side) $(Z_{eq})_{HV} = \frac{60}{3} = 20 \Omega \Rightarrow (X_{eq})_{HV} = \sqrt{20^2 - 5^2} = 19.36 \Omega$

$$\text{Rated current on the HV side} = \frac{10,000}{2500} = 4 \text{ A}$$

$$\text{Loss in windings at full load} = 45 \times \left(\frac{4}{3}\right)^2 = 80 \text{ W}$$

$$\text{Core loss (from open-circuit test)} = 50 \text{ W}$$

$$(\eta)_{1/4 \text{ load, } 0.8 \text{ p.f.}} = \frac{10,000 \times \left(\frac{1}{4}\right) \times 0.8 \times 100}{10,000 \times \left(\frac{1}{4}\right) \times 0.8 + 50 + 80 \left(\frac{1}{4}\right)^2} = 97.32\%$$

Answer: $\eta = 97.32\%$

- (b) Calculate the load (kVA output) at which maximum efficiency occurs. Also calculate the value of the maximum efficiency η_{max} at a lagging power factor of 0.8.

Maximum efficiency occurs when $P_{core} = P_{windings}$

$$\Rightarrow x^2 \times 80 = 50 \Rightarrow x = 0.79$$

$$\therefore \text{Maximum efficiency occurs at } 0.79 \times 10,000 \text{ VA} = 7.9 \text{ kVA}$$

$$\eta_{max}|_{0.8 \text{ p.f.}} = \frac{7900 \times 0.8}{7900 \times 0.8 + 50 + 50} \times 100 = 98.44\%$$

Answers: load (kVA) = 7.9 $\left[\begin{smallmatrix} 7900 \text{ VA or} \\ 7.9 \text{ kVA} \end{smallmatrix} \right]$ $\eta_{max} = 98.44\%$

All the best!